

2.2 The limit of a function.

Intro:

$$\text{let } f(x) = \frac{e^x - 1}{x} \quad ; \quad Df = \mathbb{R} \setminus \{0\}$$

$$f(1) = e - 1 \quad \text{since } 1 \in Df$$

* Can you calculate $f(0)$??

No, you cannot because $0 \notin Df$

* What happens to the value $f(x)$ if x gets "closer and closer to 0"?

(refer to unit 2.1)

$f(x)$ gets closer and closer to 1

Therefore I can say: "As x tends to 0, $f(x)$ tends to 1" and write

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

"the limit of $f(x)$ as x tends to 0 is equal to 1"

Definition 1 : Suppose $f(x)$ is defined when x is near the number a . (This means that f is defined on some open interval that contains a , except possibly at a itself.)

Then we write

$$\lim_{x \rightarrow a} f(x) = L$$

and say "the limit of $f(x)$, as x approaches a , equals L "

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

graph of $f(x) = \frac{e^x - 1}{x}$

